

Final Project Proposal

Financial Earnings Call Analyzer — A RAG-Powered Research Assistant

Due: May 14, 2026

1. Group Members

Name	Role
Harshil Chetankumar Modh	RAG Prompt Design & OpenAI API Integration
Krish Tarkar	Data Ingestion & Chunking Pipeline
Dev Desai	Vector DB & Embedding Integration
Bhuvan Chandra Alladi	Evaluation Harness & Streamlit UI

2. Business Problem

Financial analysts, portfolio managers, and institutional investors are required to review large volumes of SEC filings — including 10-Q quarterly reports and 10-K annual reports — as well as earnings call transcripts, in order to extract actionable insights. This process is highly time-consuming and error-prone when performed manually.

Our project addresses this problem by building a **Retrieval-Augmented Generation (RAG)** system that allows users to query a curated corpus of financial filings using natural language. Instead of reading hundreds of pages, an analyst can ask questions such as:

- "How did Apple's gross margin trend across Q1–Q3 2024?"
- "What risks did Microsoft cite in their most recent 10-K?"
- "Compare Google and Microsoft's R&D spending as a percentage of revenue."
- "What forward guidance did Apple give for the next quarter?"

The system retrieves the most relevant document chunks and uses OpenAI's GPT models to synthesize a cited, accurate answer — dramatically reducing research time and improving consistency of financial analysis.

3. Documents and Data Files

The project will use publicly available financial documents sourced from **SEC EDGAR** (Electronic Data Gathering, Analysis, and Retrieval system), which provides free access to all U.S. public company

filings at data.sec.gov.

Document Corpus

Company	Ticker	Filing Types	# Documents
Apple Inc.	AAPL	10-Q (Q1–Q3 2024) + 10-K (FY2024)	4
Microsoft Corp.	MSFT	10-Q (Q1–Q3 2024) + 10-K (FY2024)	4
Alphabet Inc.	GOOGL	10-Q (Q1–Q3 2024) + 10-K (FY2024)	4
Total			12 PDFs

Additional Data Assets

- **Chunked text segments:** Each PDF is parsed and split into overlapping chunks (~500 tokens) tagged with metadata: {company, filing_type, quarter, year, section}.
- **Embeddings store:** All chunks embedded using *sentence-transformers (all-MiniLM-L6-v2)* and stored in a local ChromaDB vector database.
- **Golden Q&A evaluation set:** A manually curated JSON file of 25–30 question-answer pairs with verified ground truth answers and source quotes extracted directly from the filings.

4. Evaluation of Model Output Accuracy

Model output quality will be evaluated across three dimensions using the manually curated golden Q&A set:

Metric	Description	Method
Factual Accuracy	Does the model's answer match the verified ground truth from the filing?	Binary score (0/1) by human-reviewed Q&A pairs.
Retrieval Precision	Was the correct source chunk retrieved in the top-3 results?	Top-3 retrieved chunk metadata against source_documents.
Hallucination Rate	Did the model cite a number or fact not present in any retrieved chunk?	Manual spot-check + LLM-as-judge verification.
LLM-as-Judge Score	GPT-4 rates the model answer vs ground truth on a 1-5 scale.	Automated secondary GPT-4 review, averaged across a subset of questions.

Evaluation Targets

- Factual accuracy ≥ 80% on the 25-question golden set
- Retrieval precision (top-3 hit rate) ≥ 85%
- Hallucination rate ≤ 10%
- Average LLM-as-judge score ≥ 4.0 / 5.0

5. Technical Approach

Component	Technology
PDF Parsing	pdfplumber / pymupdf
Chunking	LangChain RecursiveCharacterTextSplitter
Embeddings	sentence-transformers (all-MiniLM-L6-v2)
Vector Database	ChromaDB (local)
LLM / Generation	OpenAI GPT (openai Python SDK)
Evaluation	Custom golden set + LLM-as-judge (GPT-4)
User Interface	Streamlit
Language	Python 3.11+

All data sourced from SEC EDGAR (data.sec.gov) — publicly available, no licensing restrictions.